

Experiment	<i>Lab 4: MWT – Forces and Moments on a Canard Delta Wing (Micaplex Wind Tunnel)</i>
Author(s)	Clancy Crawford
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Section Number	15 Pratik Deshpande

Summary

Forces and Moments on a Canard Delta Wing (Micaplex Wind Tunnel) lab (MWT) quantifies the aerodynamic forces and moments acting on a canard-delta wing model in a low-speed wind tunnel. Six force components were measured, and aerodynamic tares were removed using given image and inverted data to isolate the true aerodynamic loads. Results showed that the lift increased almost linearly at low angles before stall (around 20–25°), while drag increased slowly and then sharply at higher angles due to flow separation. The lift-to-drag ratio reached a maximum at moderate angles, with higher Reynolds numbers slightly improving aerodynamic efficiency. Moment coefficients and side forces showed increased asymmetry under yaw and stronger nonlinear effects at higher angles of attack. Overall, the results confirm expected low-speed aerodynamic trends for a canard-delta configuration and show the importance of proper tare correction and Reynolds number effects on aerodynamic performance.

Aim & Objectives

The aim of this experiment was to determine the aerodynamic forces and moments on a canard-delta wing model in a low-speed wind tunnel and evaluate angle of attack, yaw angle, and Reynolds number. Objectives include operating and understanding a closed-circuit wind tunnel, an external force balance for a canard-delta wing model, understanding aerodynamic tares for the model, calculating aerodynamic force and moment coefficients, and comparing AoA's with aerodynamic characteristics.

Uncertainty Analysis

*The method used to calculate uncertainty is from the handout given that all force balance measurements have an uncertainty of 1%, dynamic pressure and geometric reference quantities. The force balance already has gravity tares removed, while aerodynamic tares must still be corrected using the given image and inverted tare files. Force and moment coefficients were calculated with $C_F = \frac{F}{q*S}$ and $C_M = \frac{M}{q*S*\bar{c}}$ respectively. The rubric does not require error bars for force, moments, and coefficient plots. A 1% measurement uncertainty in the balance produces small uncertainty relative to the overall data, so the conclusions remain relatively unaffected. All computations were done in MATLAB (Appendix A).*

Results & Discussion

Ambient temperature at ~ 71.91° and pressure at ~ 17.74PSIa are in the data set given (section 15 yaw 10 gravity tare). Aerodynamic tare was removed using the provided documentation and given datasets: image and inverted. The method used for this can be found in the Appendix under aero tare correction.

Corrected Lift vs. Angle of Attack (AoA)

Figure 1 below shows the lift forces for yaw angles of attack ranging from -5° to 35° in 5° intervals with respective Reynolds numbers. Lift increases approximately linearly with AoA until about 20°,

this is consistent with aerodynamic theory. At higher angles of attack, the lift curves begin to curve, indicating flow separation. Uncertainty bars are shown for each measurement properly.

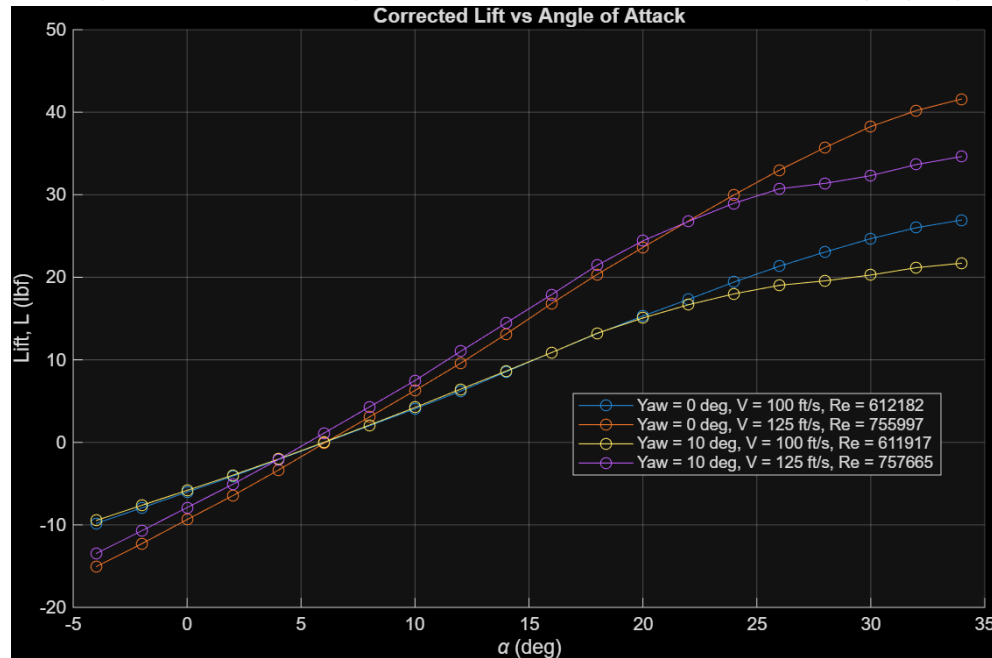


Figure 1: Corrected Lift versus Angle of Attack for two different measurements of yaw and velocity.

Corrected Drag vs. Angle of Attack

Figure 2 below showcases drag as a function of AoA. Drag is relatively low at small AoA's and increases slowly until about 15°, then increases faster. This faster increase can be attributed to flow separation and increase pressure drag. These results are consistent with classical aerodynamic theory.

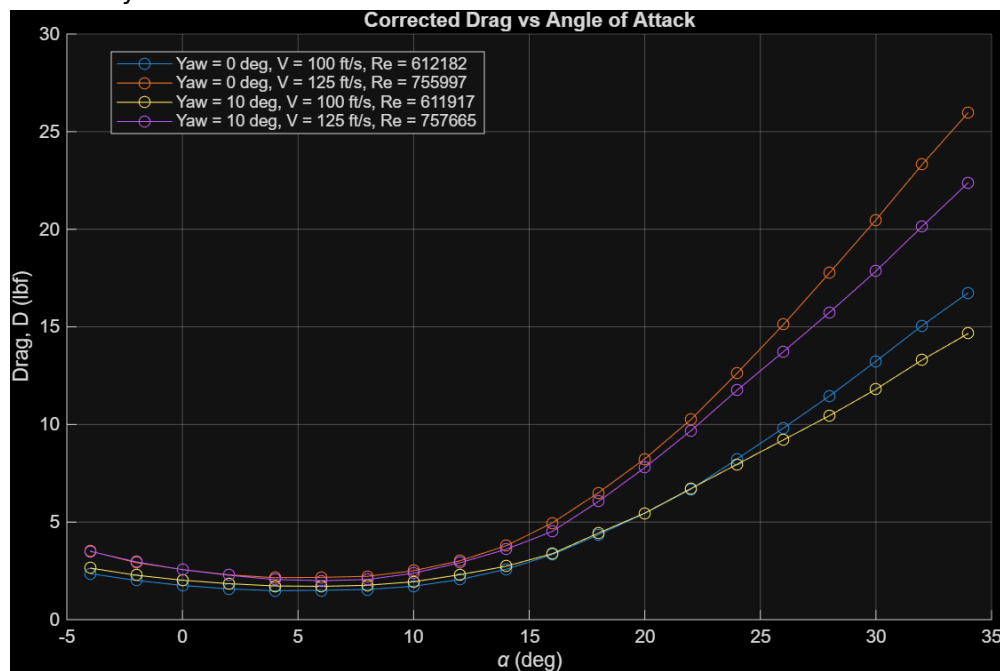


Figure 2: Corrected Drag vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Corrected Pitching Moment vs Angle of Attack

Figure 3 below is corrected pitching moment as a function of AoA. The pitching moment is increasing as AoA's increases. The graph shows a relatively linear line overall, with small deviations from individual measurements. Flow separation is also shown in this graph at higher AoA's. Yet again, these results are consistent with classical aerodynamic theory.

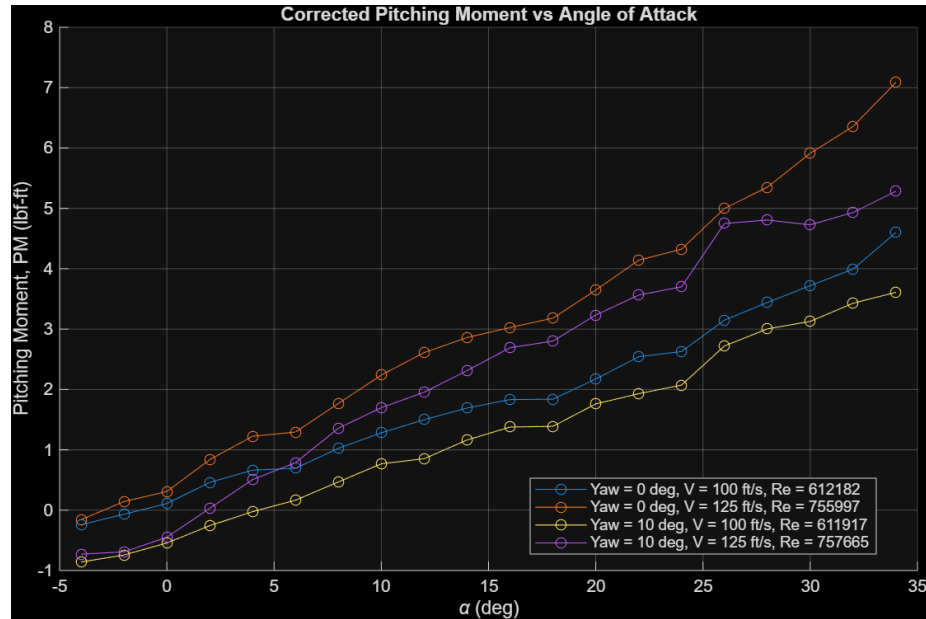


Figure 3: Corrected Pitching Moment vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Corrected Side Force vs Angle of Attack

Figure 4 below shows corrected side force as a function of AoA. For to about 15°, the figure indicates that the side force is negligible for the 0° yaw cases, due to flow symmetry. When it increases to 10° yaw, side forces are higher and decrease with AoA due to asymmetry. After about 25°, all yaw degrees increase, this is due to flow separation, consistent with classical aerodynamic theory.

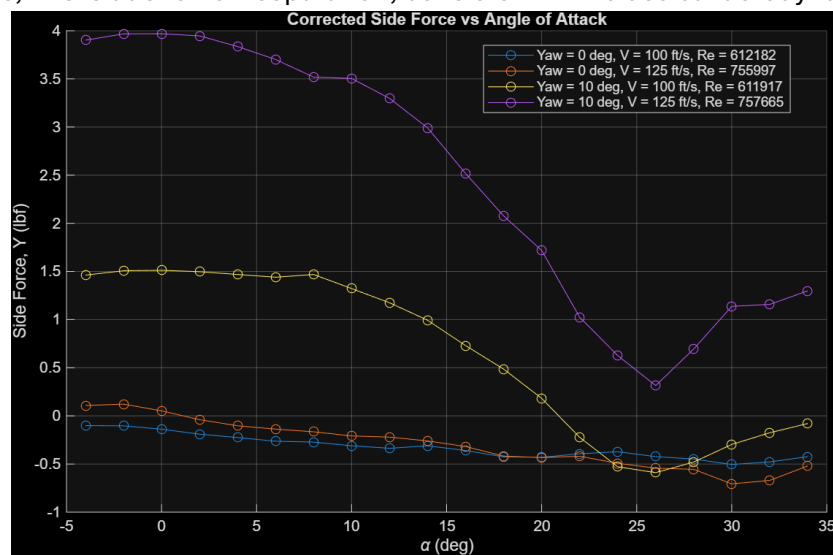


Figure 4: Corrected Side Force vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Corrected Rolling Moment vs Angle of Attack

Figure 5 below is corrected rolling moment as a function of AoA for all cases. At 0° yaw, the rolling moment remains close to zero due to symmetry. At 10° yaw, rolling moment increases linearly with angle of attack until about 20° , reflecting asymmetric loading across the wings. The increase becomes more pronounced at higher angles of attack, indicating increased instability and flow separation effects. The lift then decreases for 10° yaw around 20° , showing flow separation. The pitching moment is increasing as AoA's increases.

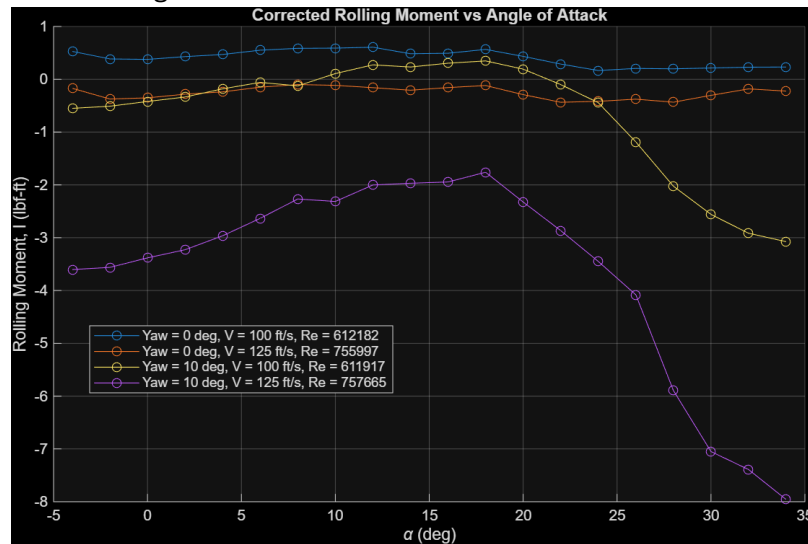


Figure 5: Corrected Rolling Moment vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Corrected Yawing Moment vs Angle of Attack

Figure 6 below is corrected yawing moment as a function of AoA for all cases. At 0° yaw, the moment remains close to zero due to symmetry. The yawing moment increases with angle of attack, particularly at higher angles because of nonlinear effects. This behavior is consistent with increased aerodynamic asymmetry and crossflow at nonzero yaw angles.

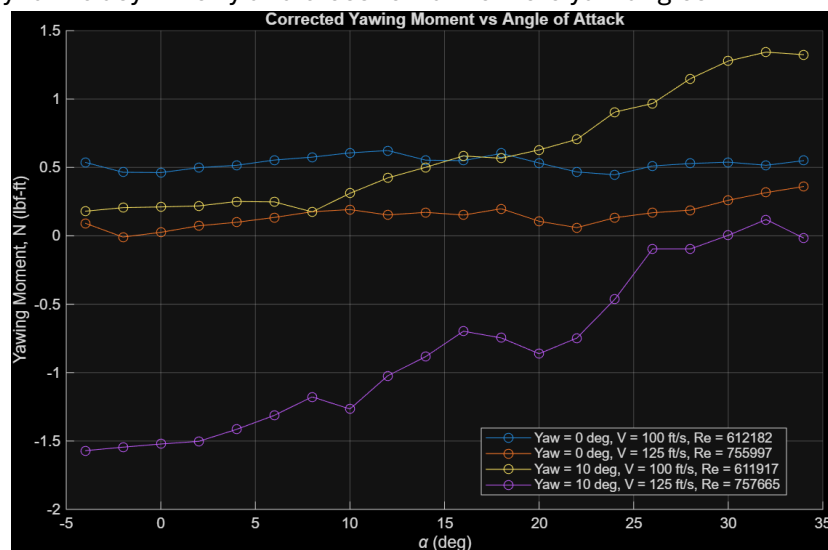


Figure 6: Corrected Yawing Moment vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Lift Coefficient vs Angle of Attack

Figure 7 below is the lift coefficient as a function of AoA for all cases. The coefficient increases linearly for all cases until about 22.5° , as consistent with airfoil theory. Above this angle of attack, the curve begins to flatten out, showing stall and flow separation. The Reynolds number has a small effect on lift. Higher yaw also reduces the amount of lift near stall due to asymmetric flow.

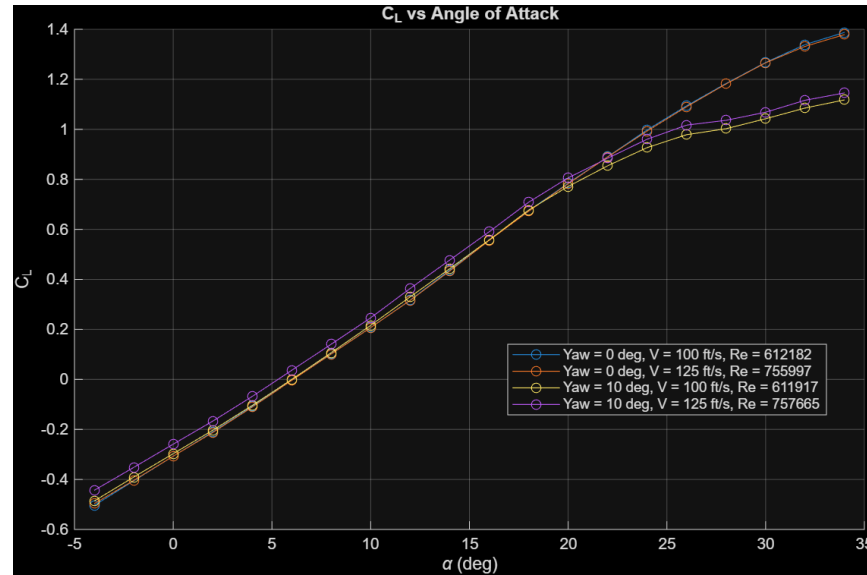


Figure 7: Coefficient of Lift vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Drag Coefficient vs Angle of Attack

Figure 8 below is the lift coefficient as a function of AoA for all cases. The coefficient is low at AoA's below $\sim 15^\circ$ but rises linearly and sharply above this threshold for all cases, as consistent with airfoil theory. At and above $\sim 25^\circ$, the curves show stall and flow separation for higher yaw reducing the amount of drag due to asymmetric flow. The Reynolds number has a minimal effect on drag.

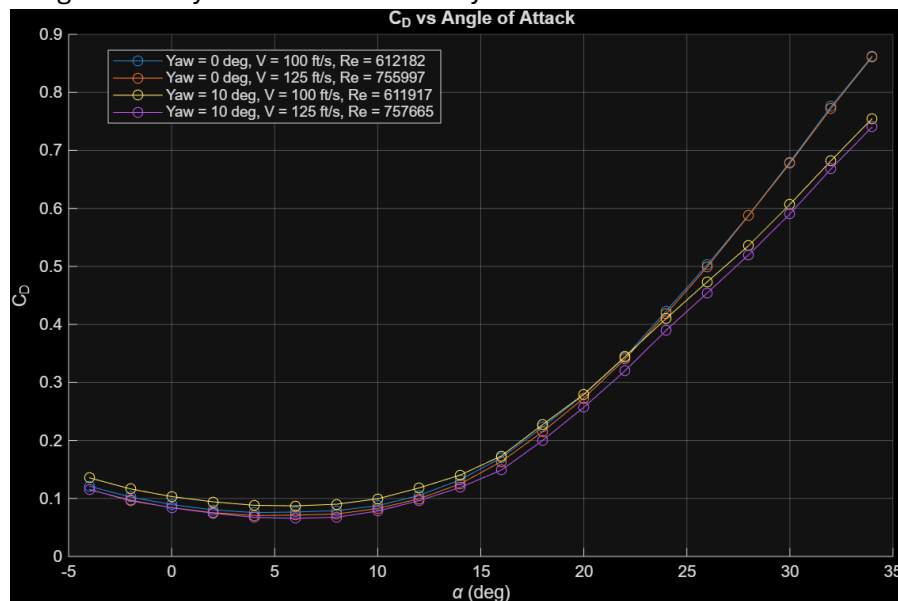


Figure 8: Coefficient of Drag vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Drag Polar (C_L vs. C_D)

Figure 9 below shows lift coefficient versus drag coefficient for all cases. At small drag coefficients ($< \sim 0.4 C_D$), the relationship is parabolic. At higher drag coefficients ($> \sim 0.4 C_D$), the curve begins to deviate from this relationship due to flow separation. Higher yaw cases also reduce the amount of lift per drag, displaying better aerodynamic efficiency.

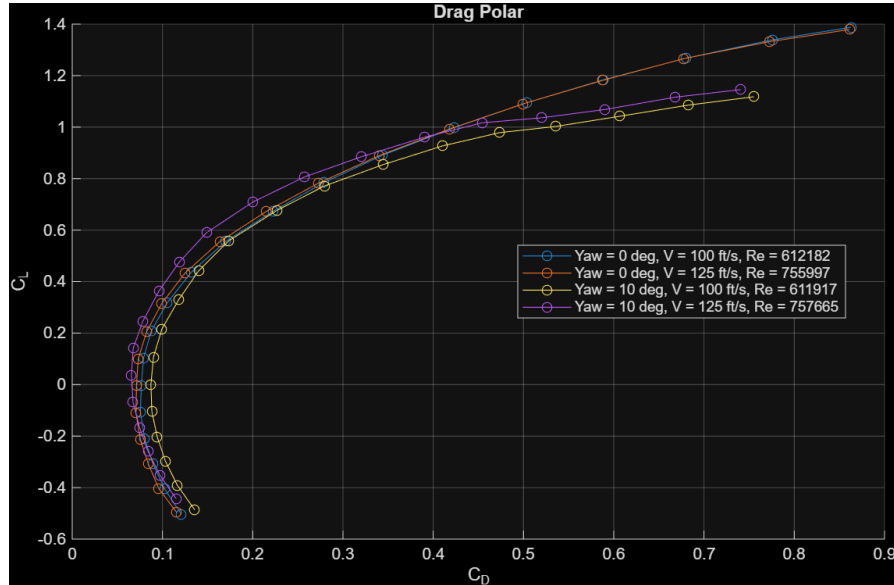


Figure 9: Drag Polar for two different measurements of yaw and velocity with Reynolds numbers.

C_L/C_D vs. Angle of Attack

Figure 10 below is the lift to drag coefficient ratio as a function of AoA for all cases. The ratio increases moderately until $\sim 15^\circ$ then decreases slowly with AoA for all cases. The maximum represents the most efficient operating conditions. Higher Reynolds numbers show higher ratio values. Higher yaw also decreases with AoA faster than a zero value for yaw.

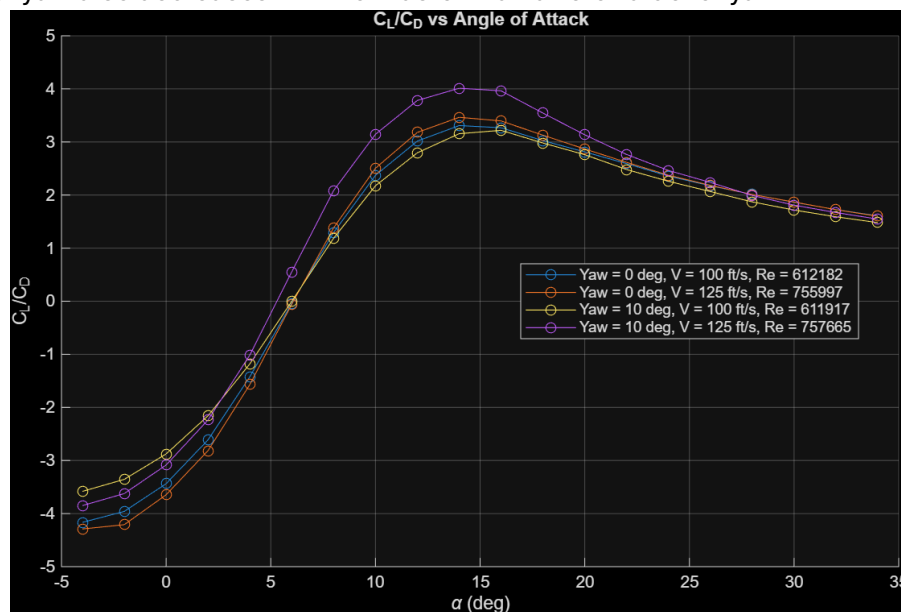


Figure 10: Lift-to-Drag ratio vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Best C_L/C_D Table for Angle of Attack

Table 1 below summarizes the angle of attack that corresponds to the maximum C_L/C_D for each velocity and yaw setting. The optimal angle of attack is the same for three cases and varies slightly for 10° yaw, 100 ft/s. Lower velocity also shows higher mean Reynolds number and lower ratio of coefficients.

Table 1: The angle of attack for best C_L/C_D at each Reynolds number and yaw angle.

Yaw (degrees)	V (ft/s)	Mean Reynolds Number	Best Alpha	Best C_L/C_D
0	100	6.1218e+05	14	3.3086
0	125	7.56e+05	14	3.4644
10	100	6.1192e+05	16	3.219
10	125	7.5767e+05	14	4.0122

Pitching Moment Coefficient vs. Angle of Attack

Figure 11 below displays the pitching moment coefficient as a function of AoA for all cases. The coefficient increases moderately linearly with angle of attack until ~25° then varies due to flow separation for all cases. This indicates a nose-up tendency. Lower yaw values show higher coefficient results for all angles of attack degrees. Reynolds number does not seem to impact this plot.

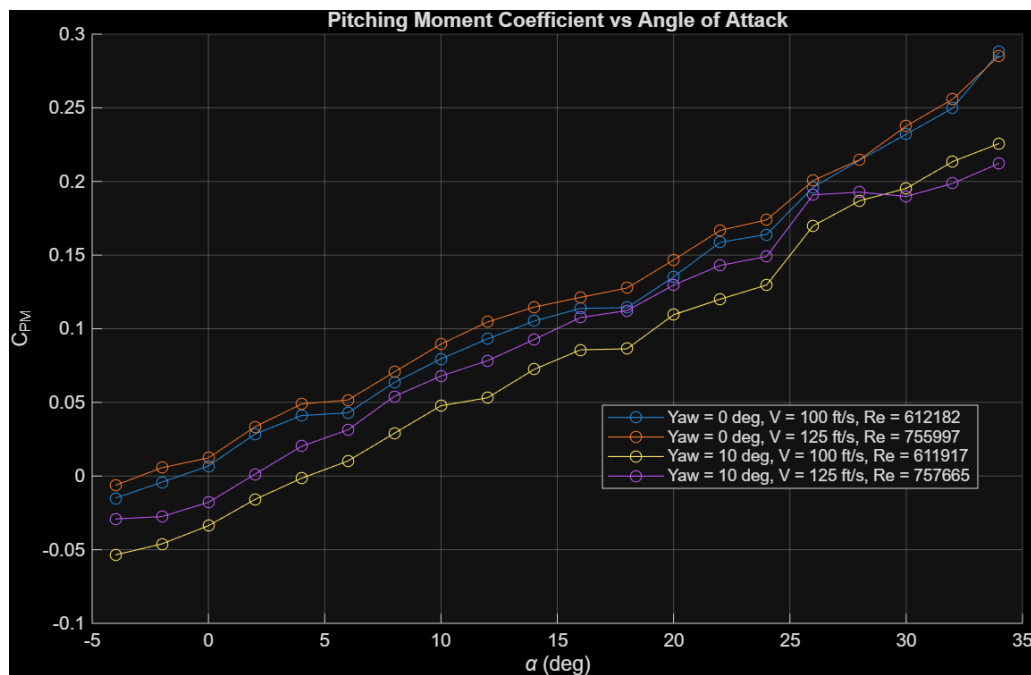


Figure 11: Coefficient of pitching moment vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Figure 12-14 below displays the variation of C_Y , C_{MX} , and C_{MZ} with angle of attack for all cases of yaw and velocity. For 0° yaw, the coefficients are close to zero along all angles of attack. At 10° yaw, the coefficient of side force is moderately constant until 10° then decreases linearly. The coefficient of rolling moment slowly and linearly increases until flow separation properties are found at ~17.5°. For the yawing moment coefficient, yaw of 10° slowly and linearly increases until flow separation. Due to flow separation, for all cases after 25° the coefficients increase/decrease respectively

based on their previous behavior. Additionally, the higher the yaw, the more pronounced the value of coefficient is, relative to each graph. Reynolds number did not seem to impact these plots.

Side Force Coefficient vs. Angle of Attack

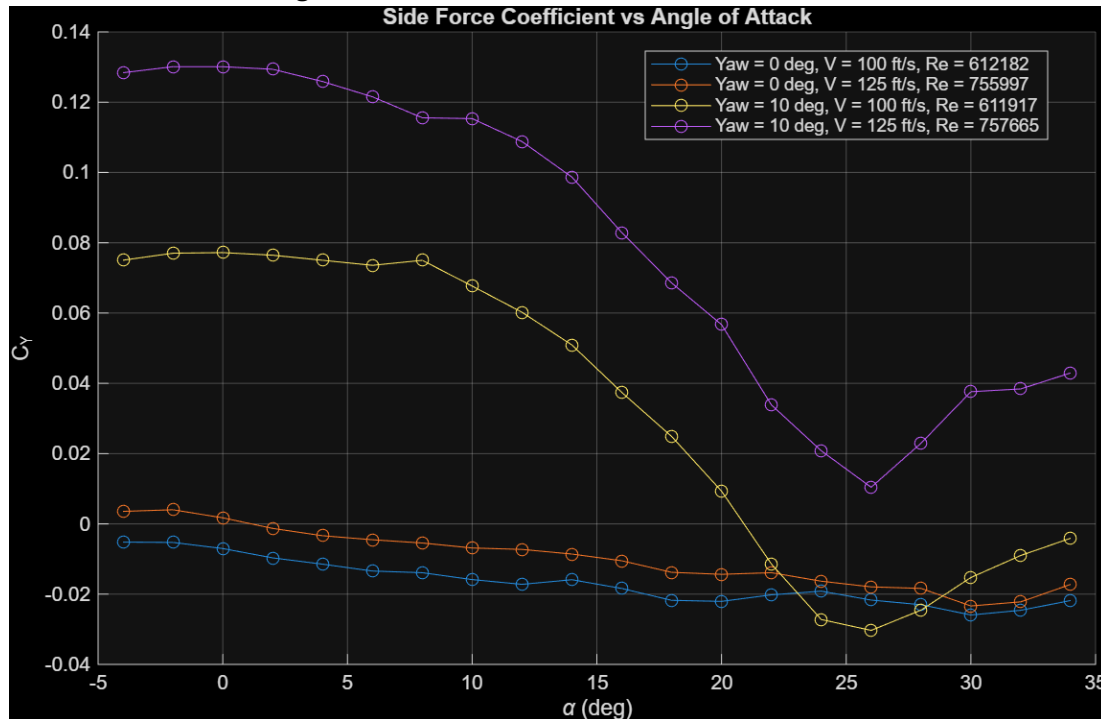


Figure 12: Coefficient of side force vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Rolling Moment Coefficient vs. Angle of Attack

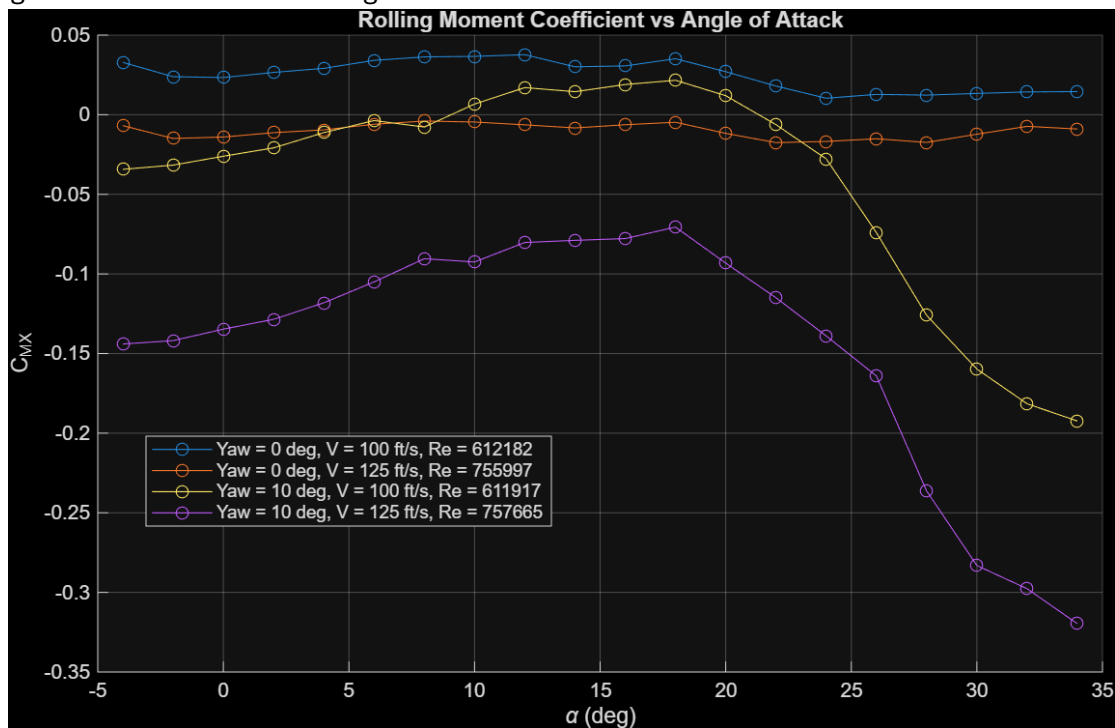


Figure 13: Coefficient of rolling moment vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

Yawing Moment Coefficient vs. Angle of Attack

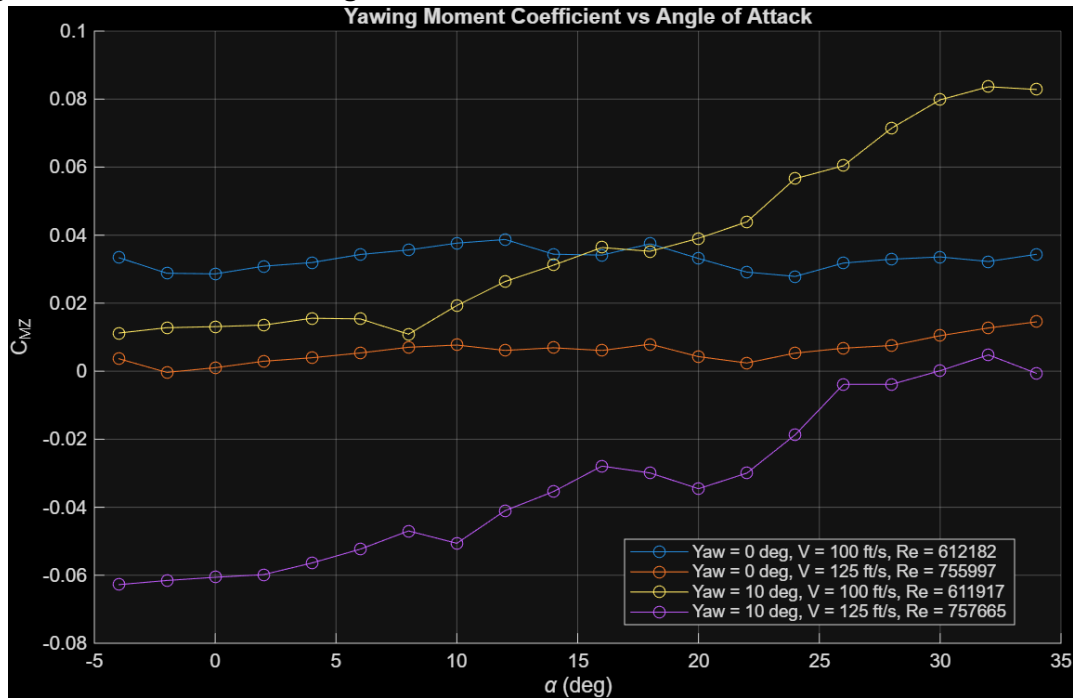


Figure 14: Coefficient of yawing moment vs. Angle of Attack with Reynolds numbers for two different measurements of yaw and velocity.

References

AE315_MWT (03.09.26).pdf

ExpAero_WindTunnels.pdf

Appendix

```
% AE315 DB15 Group 15.2: Lab 4
% Clancy Crawford Student ID:2654296
% 3/24/2026

clear;
close all;
clc;

% Givens
Sw_MW = 242.01; % in^2
cbarw_MW = 9.88; % in

Sw_MW = Sw_MW/144; % ft^2
cbar_w_MW = cbarw_MW/12; % ft
```

```

% Read measured files
meas{1} = readtable('031026_101217_sect17_Yaw0_100fts.csv','VariableNamingRule','preserve');
meas{2} = readtable('031026_102435_sect17_Yaw0_125fts.csv','VariableNamingRule','preserve');
meas{3} = readtable('031026_090533_sect15_yaw10_100fts.csv','VariableNamingRule','preserve');
meas{4} = readtable('031026_091813_sect15_yaw10_125fts.csv','VariableNamingRule','preserve');

% Read aerodynamic tare files
img{1} = readtable('Image_Yaw0_100fps.csv','VariableNamingRule','preserve');
img{2} = readtable('Image_Yaw0_125fps.csv','VariableNamingRule','preserve');
img{3} = readtable('Image_Yaw10_100fps.csv','VariableNamingRule','preserve');
img{4} = readtable('Image_Yaw10_125fps.csv','VariableNamingRule','preserve');

inv{1} = readtable('Invert_Yaw0_100fps.csv','VariableNamingRule','preserve');
inv{2} = readtable('Invert_Yaw0_125fps.csv','VariableNamingRule','preserve');
inv{3} = readtable('Invert_Yaw10_100fps.csv','VariableNamingRule','preserve');
inv{4} = readtable('Invert_Yaw10_125fps.csv','VariableNamingRule','preserve');

% setting up arrays
nCases = 4;
labels = {'Yaw 0, 100 ft/s','Yaw 0, 125 ft/s','Yaw 10, 100 ft/s','Yaw 10, 125 ft/s'};
Yaw_vals = [0; 0; 10; 10];
V_vals = [100; 125; 100; 125];

alpha = cell(1,nCases);
q = cell(1,nCases);
Re = cell(1,nCases);
D = cell(1,nCases);
Y = cell(1,nCases);
L = cell(1,nCases);
RM = cell(1,nCases);
PM = cell(1,nCases);
YM = cell(1,nCases);

D_img = cell(1,nCases);
Y_img = cell(1,nCases);
L_img = cell(1,nCases);
RM_img = cell(1,nCases);
PM_img = cell(1,nCases);
YM_img = cell(1,nCases);
D_inv = cell(1,nCases);
Y_inv = cell(1,nCases);
L_inv = cell(1,nCases);
RM_inv = cell(1,nCases);
PM_inv = cell(1,nCases);
YM_inv = cell(1,nCases);

Dc = cell(1,nCases);
Yc = cell(1,nCases);
Lc = cell(1,nCases);
RMC = cell(1,nCases);
PMC = cell(1,nCases);
YMC = cell(1,nCases);

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CD = cell(1,nCases);
CY = cell(1,nCases);
CL = cell(1,nCases);
CMX = cell(1,nCases);
CPM = cell(1,nCases);
CMZ = cell(1,nCases);
CLCD = cell(1,nCases);

mean_Re = nan(nCases,1);

% match rows by pitch using tolerance
tol = 1e-6;

for k = 1:nCases
    alpha{k} = meas{k}.("Pitch");
    q{k} = meas{k}.("Dynamic Pressure")*144;
    Re{k} = meas{k}.("Reynolds Number per ft");

    % measured forces/moments
    D{k} = meas{k}.("WAFMC Drag");
    Y{k} = meas{k}.("WAFMC Side");
    L{k} = meas{k}.("WAFMC Lift");
    RM{k} = meas{k}.("WAFMC Roll");
    PM{k} = meas{k}.("WAFMC Pitch");
    YM{k} = meas{k}.("WAFMC Yaw");

    D_img_mean = mean(img{k}.("WAFMC Drag"));
    Y_img_mean = mean(img{k}.("WAFMC Side"));
    L_img_mean = mean(img{k}.("WAFMC Lift"));
    RM_img_mean = mean(img{k}.("WAFMC Roll"));
    PM_img_mean = mean(img{k}.("WAFMC Pitch"));
    YM_img_mean = mean(img{k}.("WAFMC Yaw"));
    D_inv_mean = mean(inv{k}.("WAFMC Drag"));
    Y_inv_mean = mean(inv{k}.("WAFMC Side"));
    L_inv_mean = mean(inv{k}.("WAFMC Lift"));
    RM_inv_mean = mean(inv{k}.("WAFMC Roll"));
    PM_inv_mean = mean(inv{k}.("WAFMC Pitch"));
    YM_inv_mean = mean(inv{k}.("WAFMC Yaw"));

    % aero tare correction
    Dc{k} = D{k} - (D_img_mean - D_inv_mean);
    Yc{k} = Y{k} - (Y_img_mean - Y_inv_mean);
    Lc{k} = L{k} - (L_img_mean - L_inv_mean);
    RMc{k} = RM{k} - (RM_img_mean - RM_inv_mean);
    PMc{k} = PM{k} - (PM_img_mean - PM_inv_mean);
    YMc{k} = YM{k} - (YM_img_mean - YM_inv_mean);

    % coefficients
    CD{k} = Dc{k}/(q{k}*Sw_MW);
    CY{k} = Yc{k}/(q{k}*Sw_MW);
    CL{k} = Lc{k}/(q{k}*Sw_MW);
    CMX{k} = RMc{k}/(q{k}*Sw_MW*cbar_w_MW);

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    CPM{k} = PMc{k}./(q{k}*Sw_MW*cbar_w_MW);
    CMZ{k} = YMc{k}./(q{k}*Sw_MW*cbar_w_MW);
    CLCD{k} = CL{k}./CD{k};

    mean_Re(k) = mean(Re{k}, 'omitnan');
end

% plotting
% Legends with Re #
labels = cell(1,nCases);
for k = 1:nCases
    labels{k} = sprintf('Yaw = %d deg, V = %d ft/s, Re = %.0f', Yaw_vals(k), V_vals(k),
    mean_Re(k));
end

% Corrected forces and moments vs alpha
% Lift vs alpha
figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, Lc{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('Lift, L (lbf)')
title('Corrected Lift vs Angle of Attack')
legend('Location','best')

% Drag vs alpha
figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, Dc{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('Drag, D (lbf)')
title('Corrected Drag vs Angle of Attack')
legend('Location','best')

% Pitching moment vs alpha
figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, PMc{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('Pitching Moment, PM (lbf-ft)')
title('Corrected Pitching Moment vs Angle of Attack')
legend('Location','best')

% Side force vs alpha

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```

figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, Yc{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('Side Force, Y (lbf)')
title('Corrected Side Force vs Angle of Attack')
legend('Location','best')

% Rolling moment vs alpha
figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, RMc{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('Rolling Moment, l (lbf-ft)')
title('Corrected Rolling Moment vs Angle of Attack')
legend('Location','best')

% Yawing moment vs alpha
figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, YMc{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('Yawing Moment, N (lbf-ft)')
title('Corrected Yawing Moment vs Angle of Attack')
legend('Location','best')

% coefficients
% CL vs alpha
figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, CL{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('C_L')
title('C_L vs Angle of Attack')
legend('Location','best')

% CD vs alpha
figure
hold on;
grid on
for k = 1:nCases

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    plot(alpha{k}, CD{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('C_D')
title('C_D vs Angle of Attack')
legend('Location','best')

% Drag polar
figure
hold on;
grid on
for k = 1:nCases
    plot(CD{k}, CL{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('C_D')
ylabel('C_L')
title('Drag Polar')
legend('Location','best')

% CL/CD vs alpha
figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, CLCD{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('C_L/C_D')
title('C_L/C_D vs Angle of Attack')
legend('Location','best')

% Best CL/CD table
best_alpha = nan(nCases,1);
best_clcd = nan(nCases,1);

for k = 1:nCases
    thisCLCD = CLCD{k};
    thisAlpha = alpha{k};
    valid = isfinite(thisCLCD) & isfinite(thisAlpha);
    thisCLCD = thisCLCD(valid);
    thisAlpha = thisAlpha(valid);

    [best_clcd(k), idx] = max(thisCLCD);
    best_alpha(k) = thisAlpha(idx);
end

BestTable = table(Yaw_vals, V_vals, mean_Re, best_alpha, best_clcd, 'VariableNames',
{'Yaw_deg', 'V_fts', 'Mean_Re', 'Alpha_best_CLCD', 'Best_CLCD'});
disp(BestTable)

% Pitching moment coefficient vs alpha
figure
hold on;

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```

grid on
for k = 1:nCases
    plot(alpha{k}, CPM{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('C_{PM}')
title('Pitching Moment Coefficient vs Angle of Attack')
legend('Location','best')

% CY, CMX, CMZ vs alpha
% Side force coefficient
figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, CY{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('C_Y')
title('Side Force Coefficient vs Angle of Attack')
legend('Location','best')

% Rolling moment coefficient
figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, CMX{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('C_{MX}')
title('Rolling Moment Coefficient vs Angle of Attack')
legend('Location','best')

% Yawing moment coefficient
figure
hold on;
grid on
for k = 1:nCases
    plot(alpha{k}, CMZ{k}, 'o-', 'DisplayName', labels{k})
end
xlabel('\alpha (deg)')
ylabel('C_{MZ}')
title('Yawing Moment Coefficient vs Angle of Attack')
legend('Location','best')

```

Yaw_deg	V_fts	Mean_Re	Alpha_best_CLCD	Best_CLCD
0	100	6.1218e+05	14	3.3086
0	125	7.56e+05	14	3.4644
10	100	6.1192e+05	16	3.219

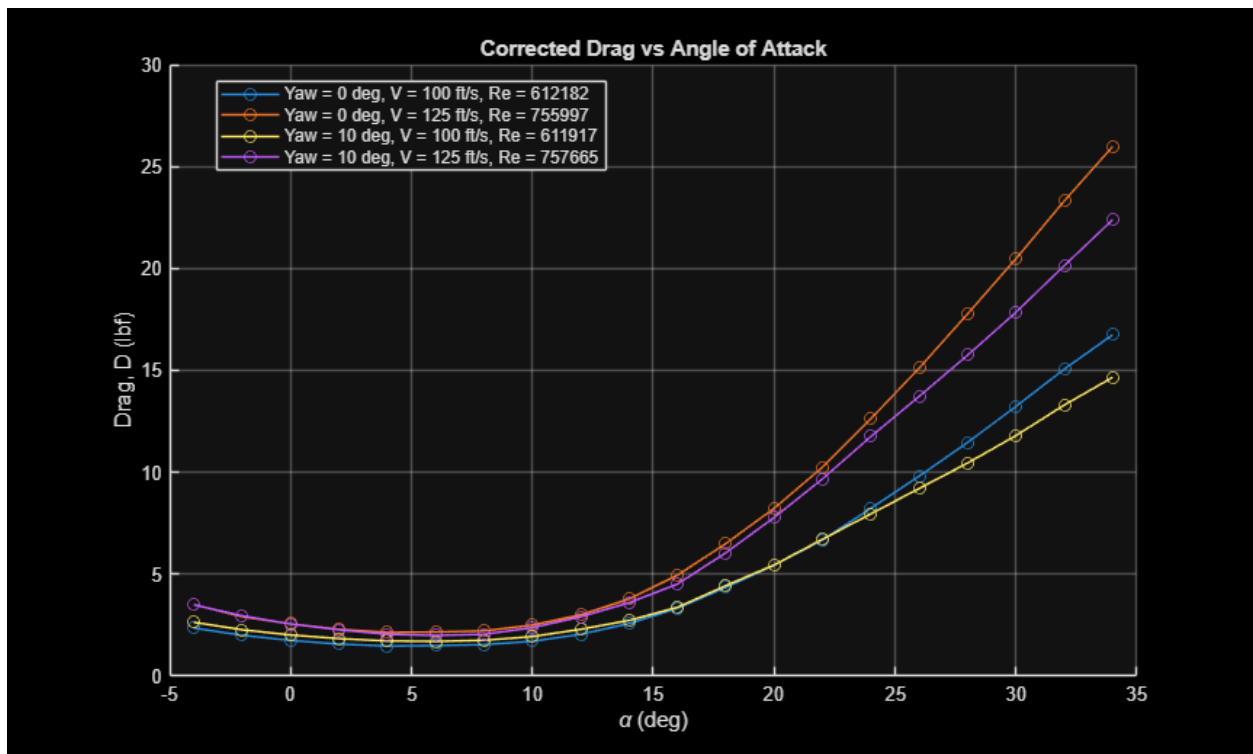
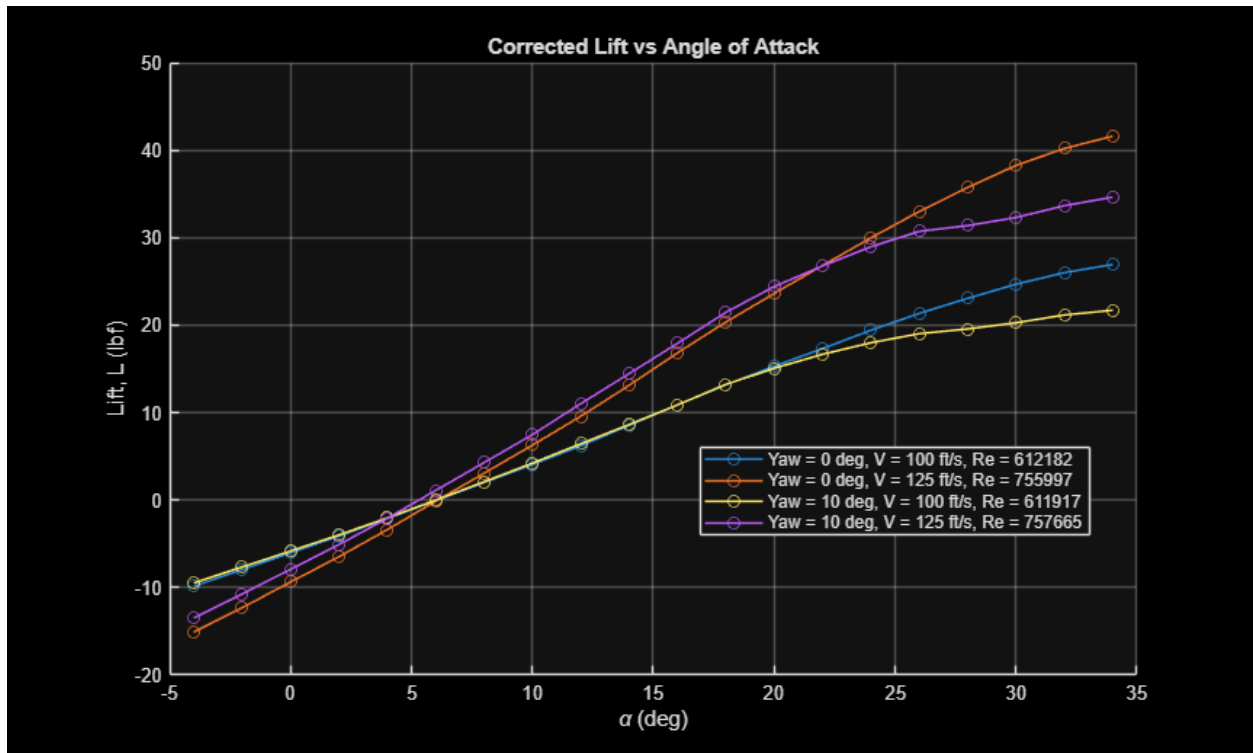
10

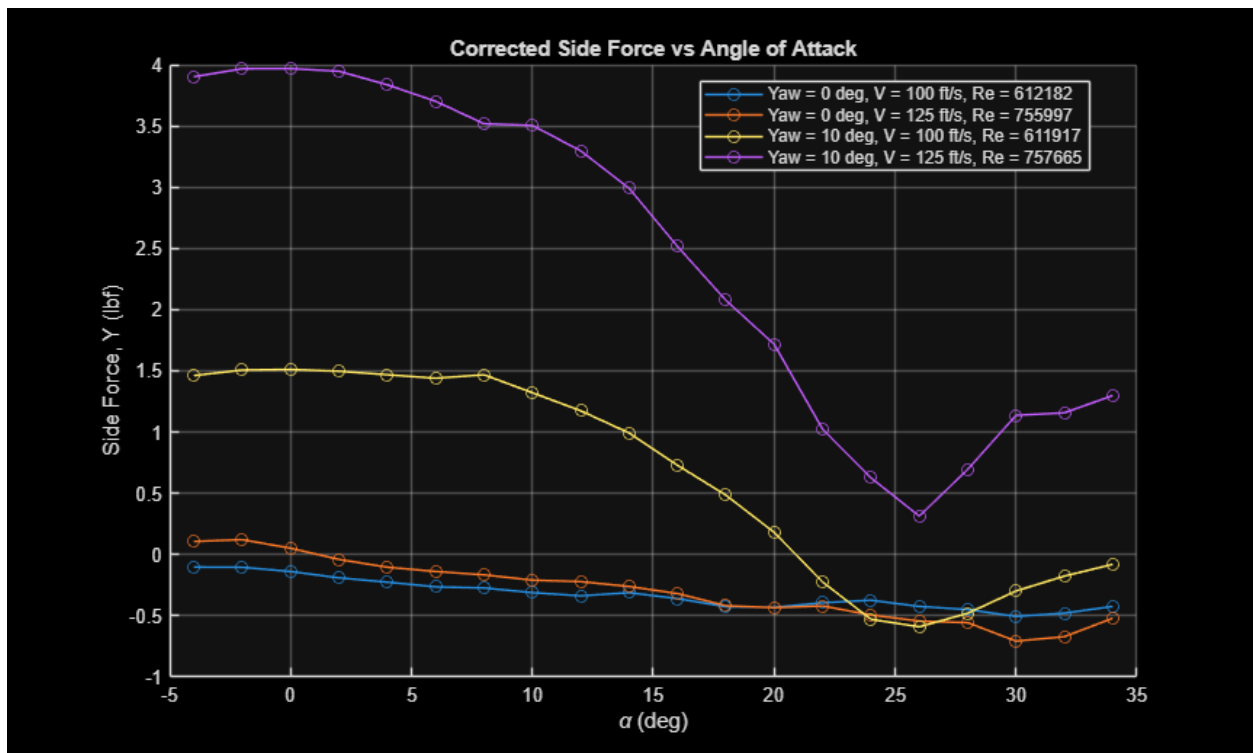
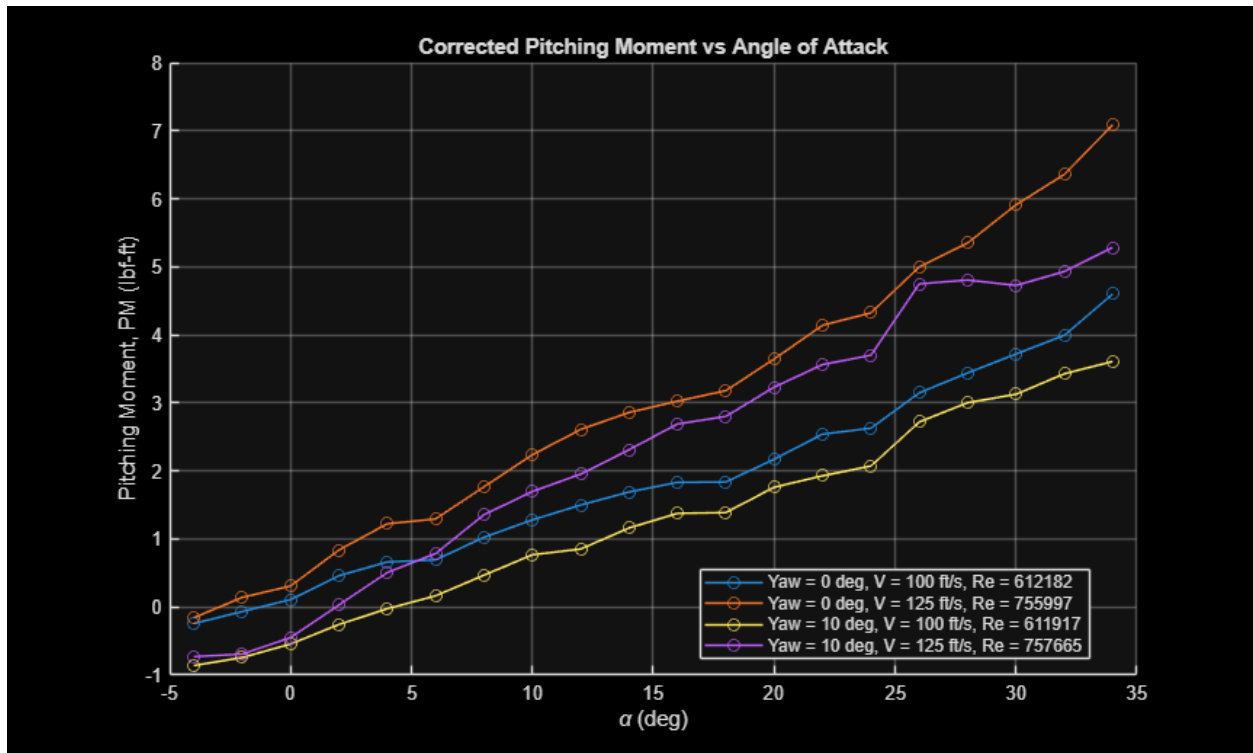
125

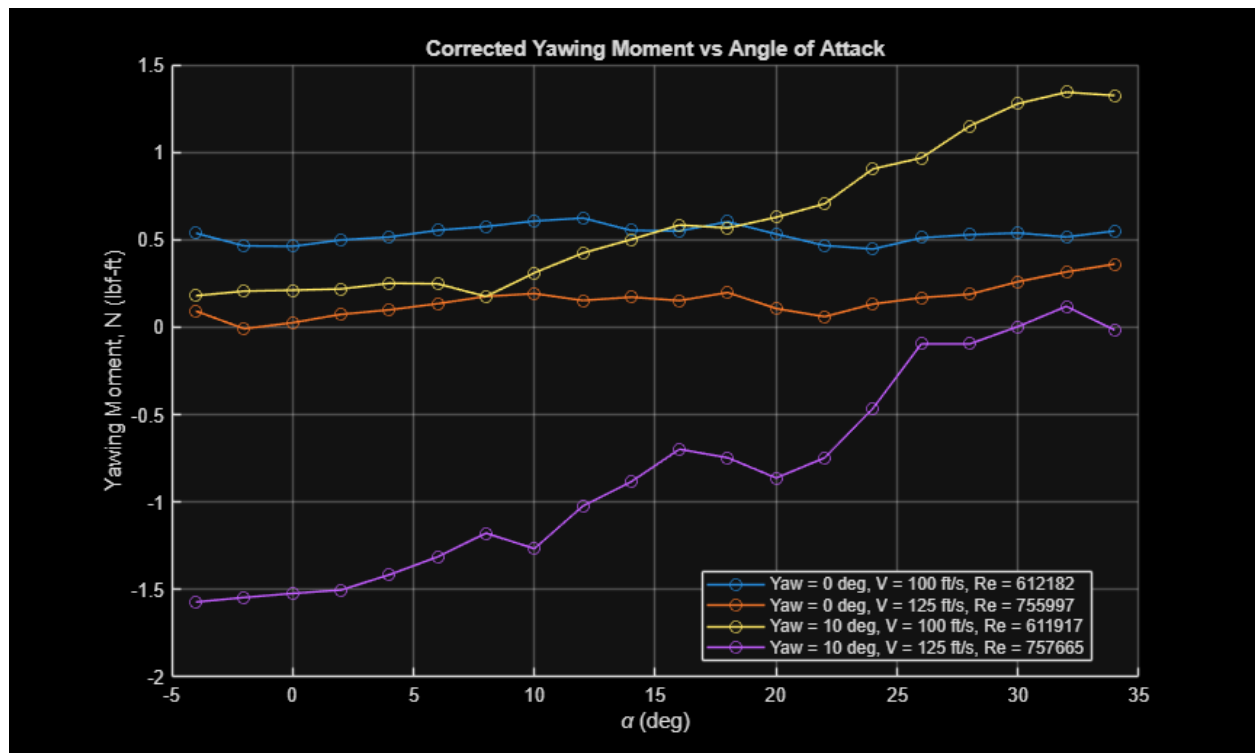
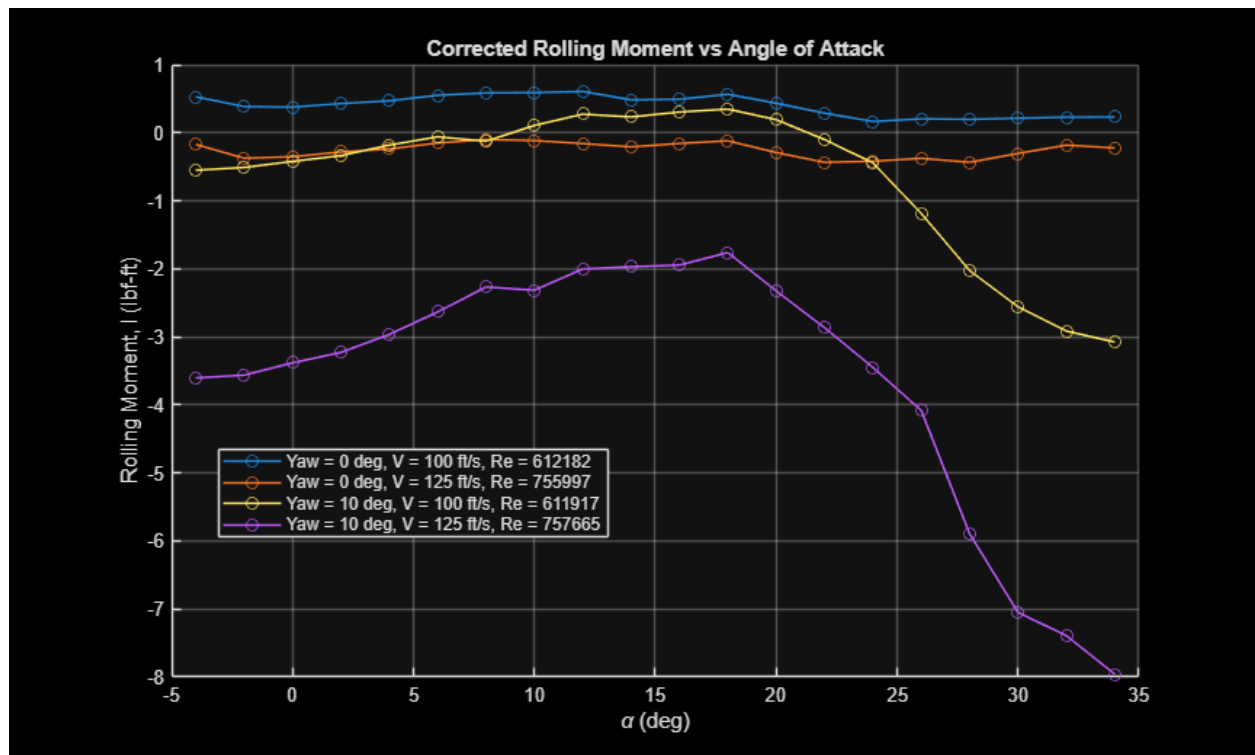
7.5767e+05

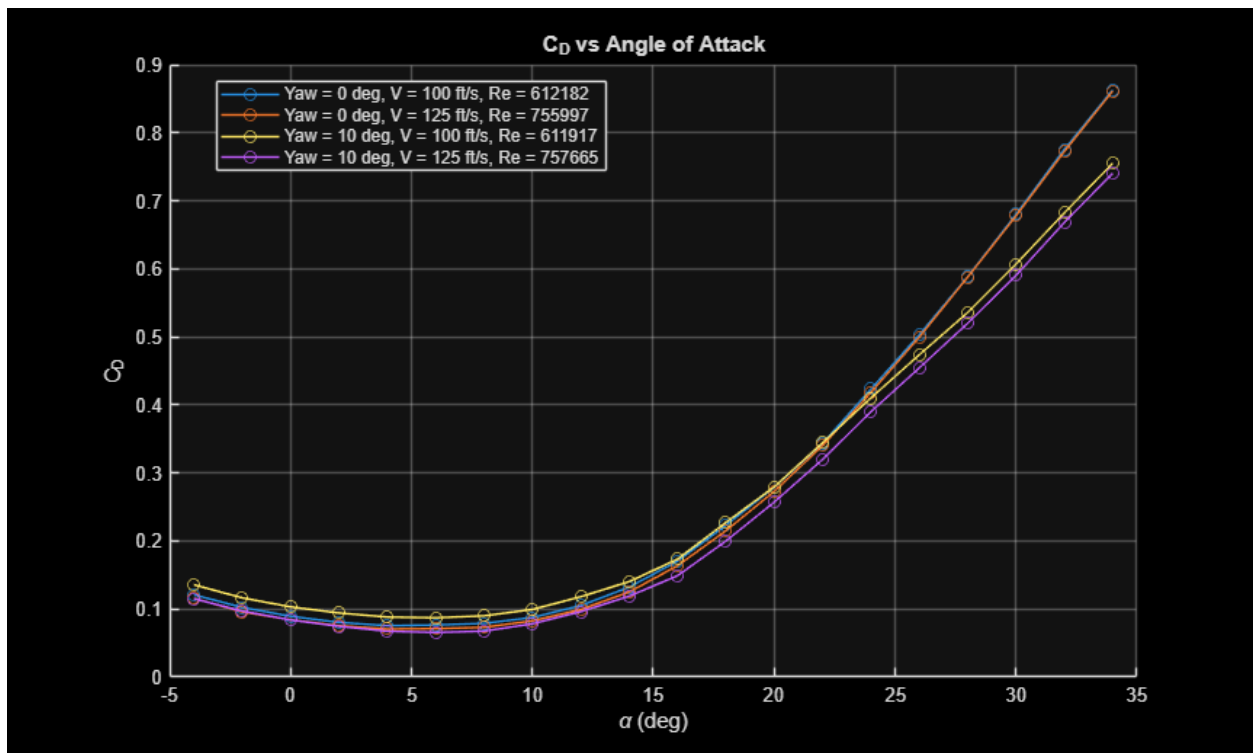
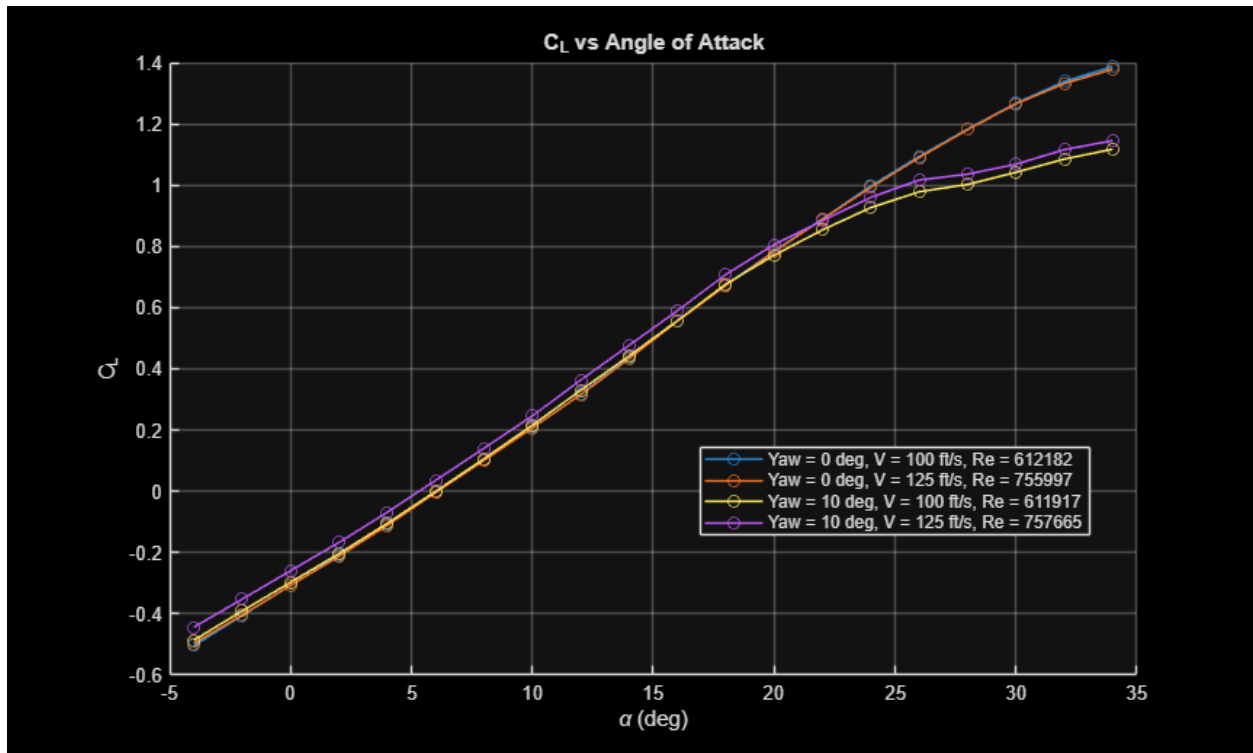
14

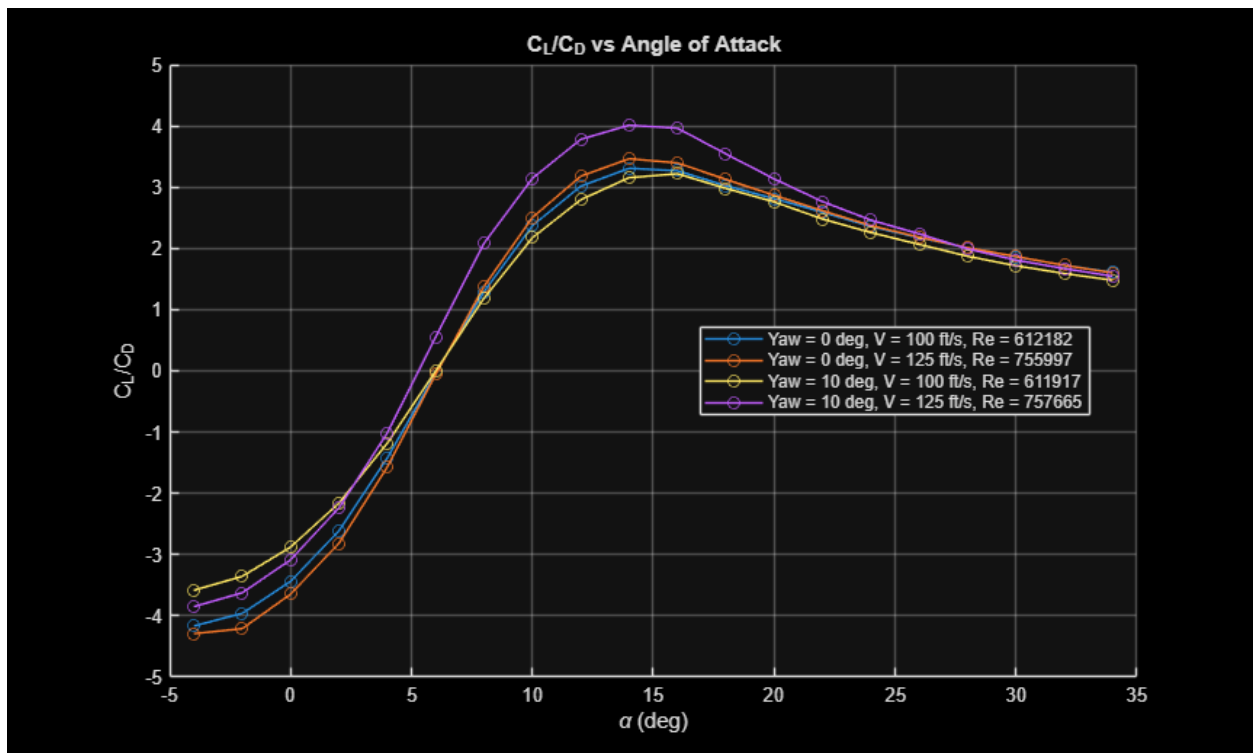
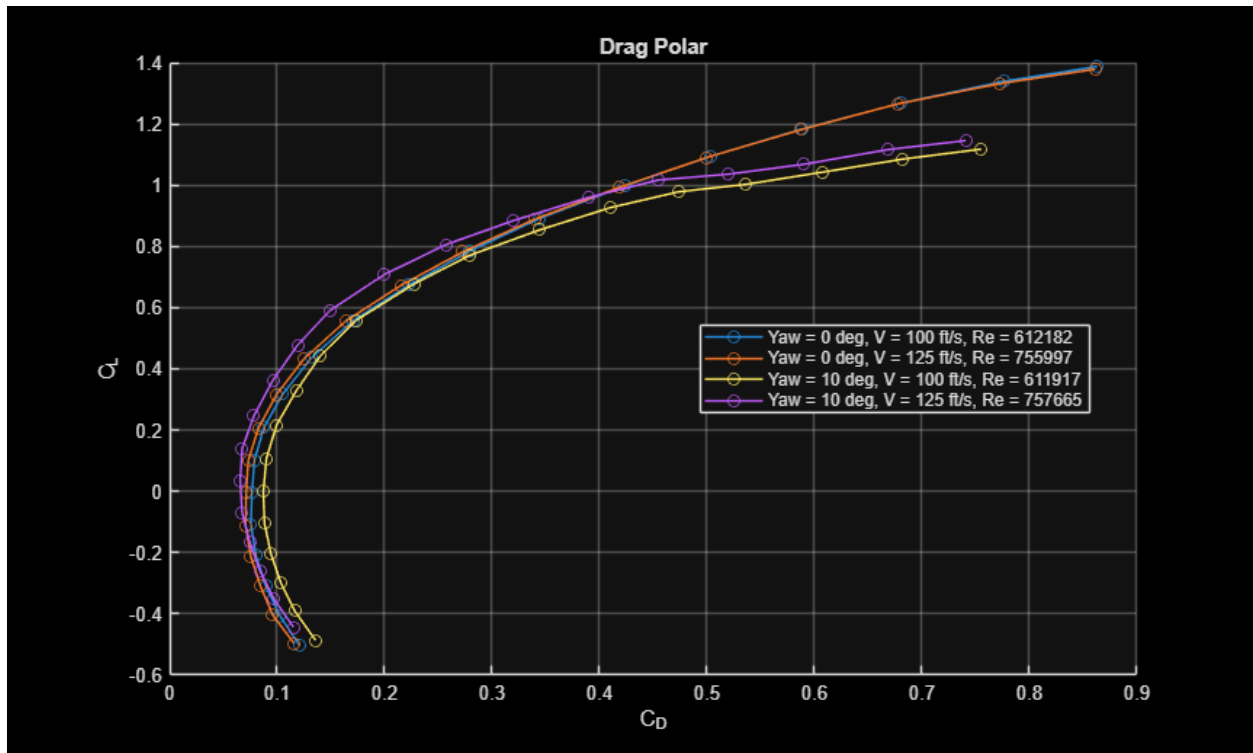
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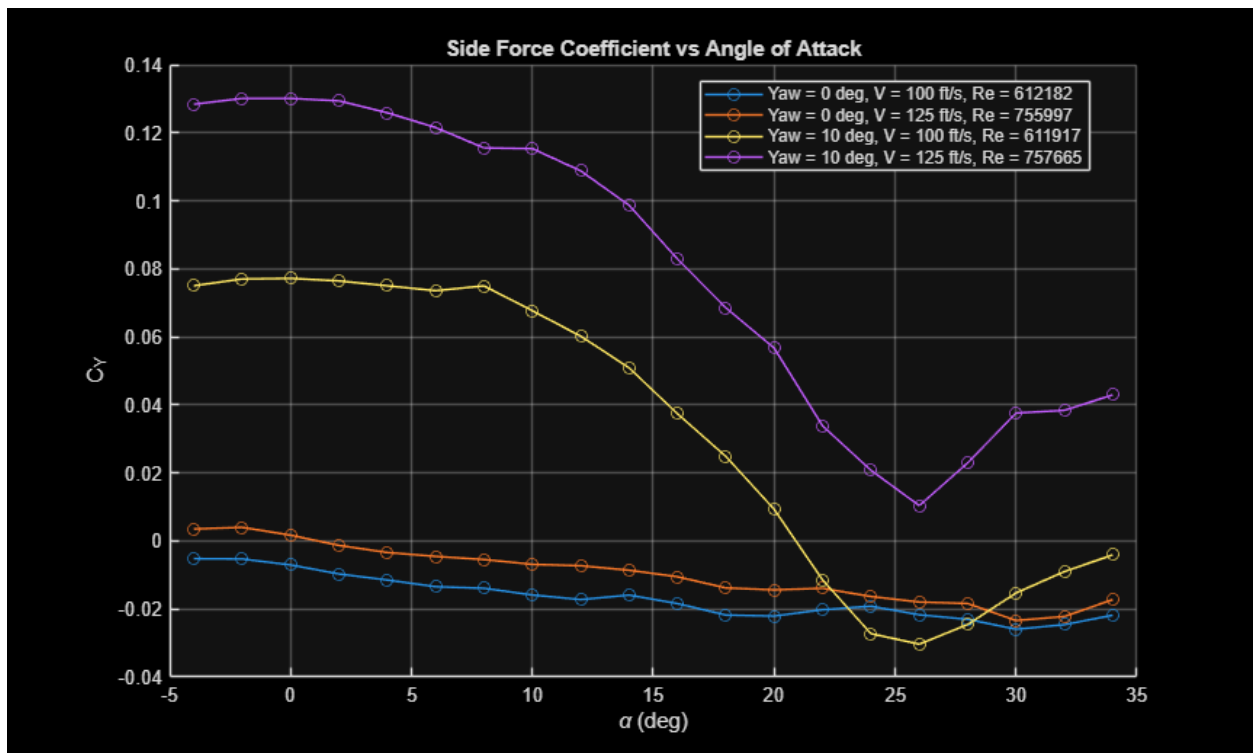
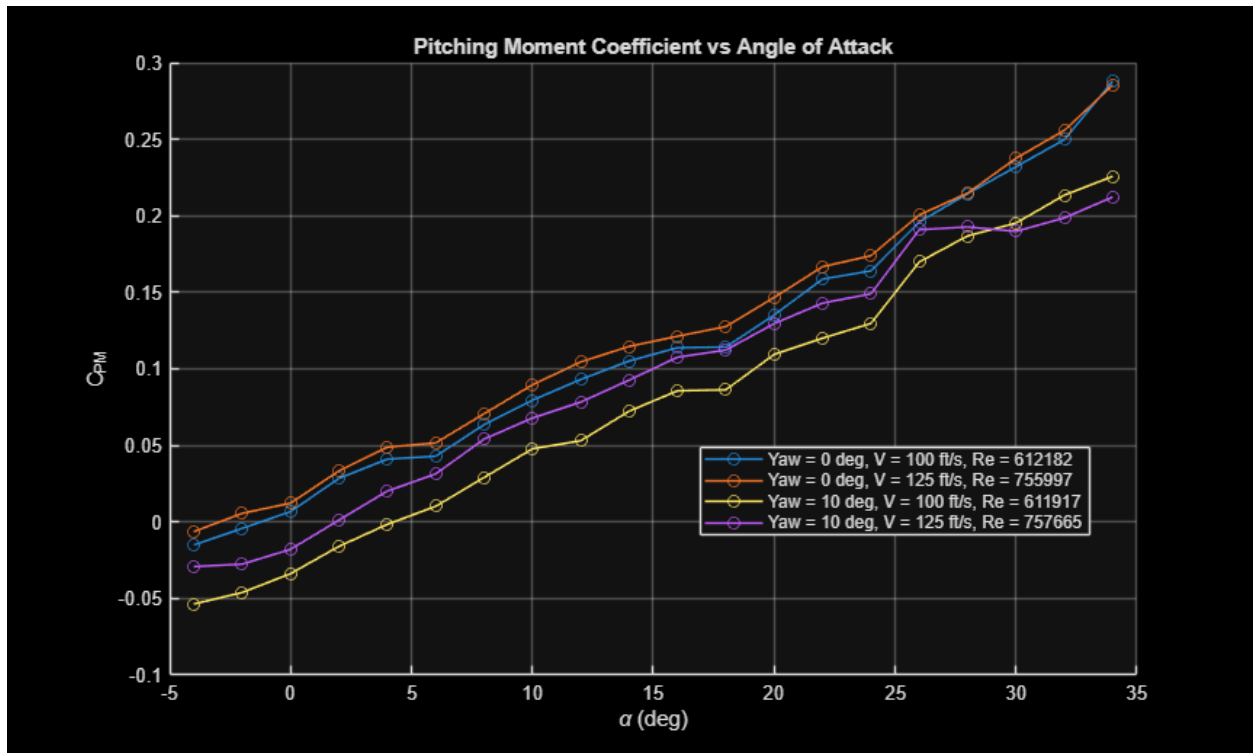


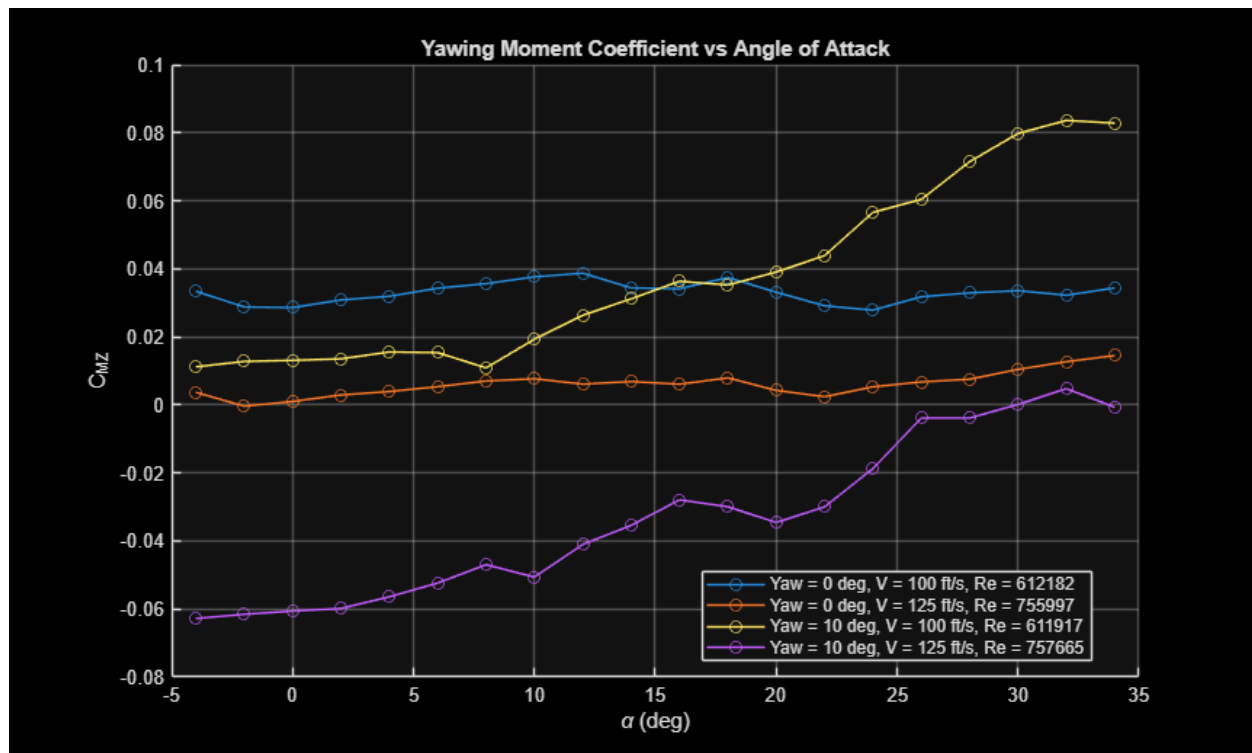
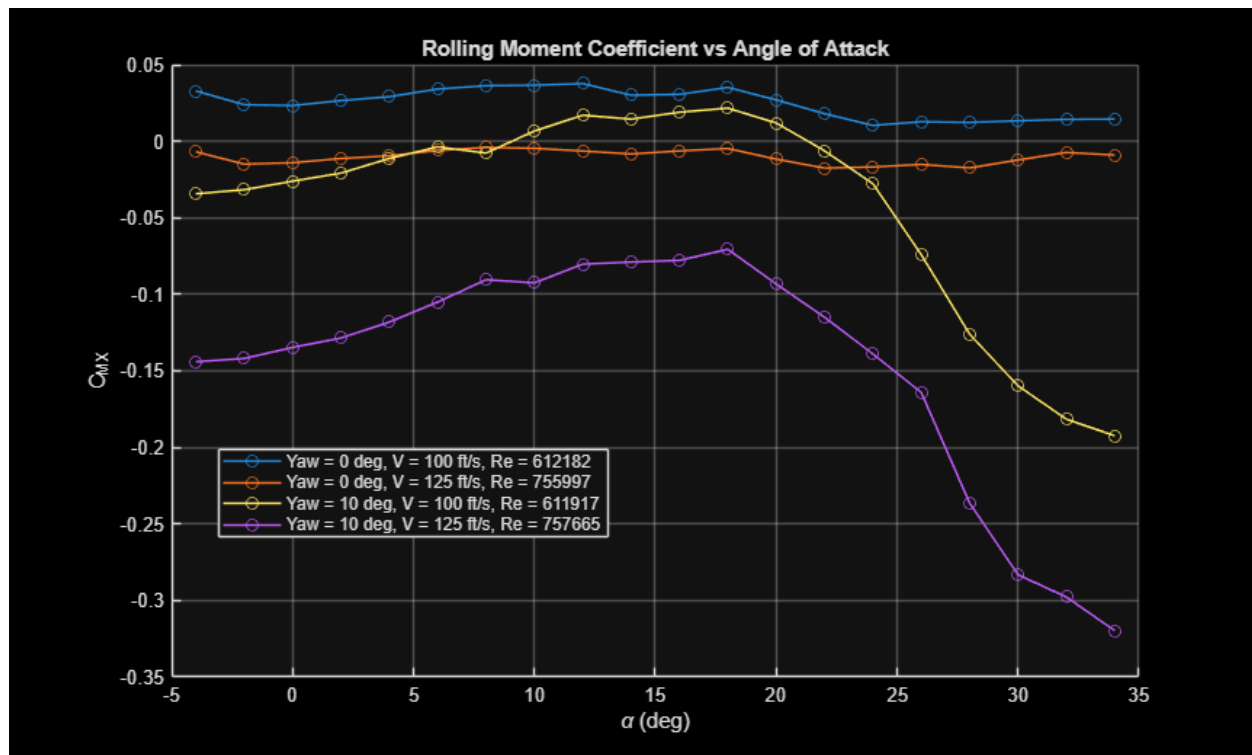












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